Chapter 24: Deployment

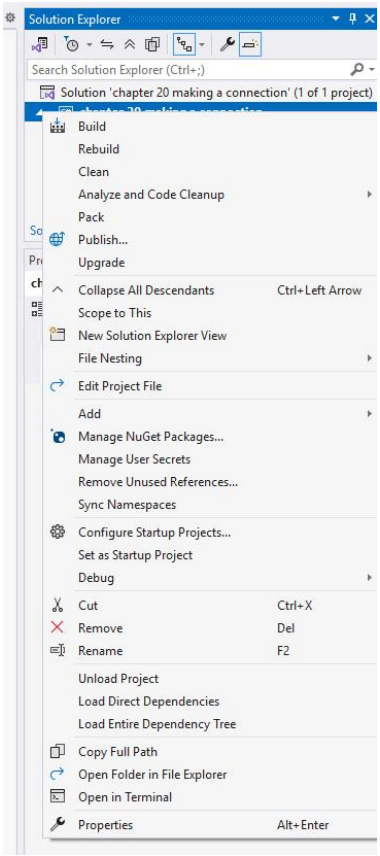
Deployment is always the very last step in the creation of an application. Unfortunately for this lesson, there are tons of different ways to deploy an application. Historically every application was loaded into IIS, Microsoft's Internet Information Service, and life was easy as your users just typed in a URL and IIS feed them the pages whether dynamic or static. Today you can deploy to IIIS, a ton of different cloud options, mobile or desktop.

So how do you choose? Well, the good news is that originally you probably will not have to as someone else is going to make those decisions. If your organization is large enough you may even have a release manager or even a release team responsible for deployment.

Regardless, let's take a quick look at Visual Studio's publish wizard and see how that guy works to give you some idea as to the options and requirements you will eventually face.

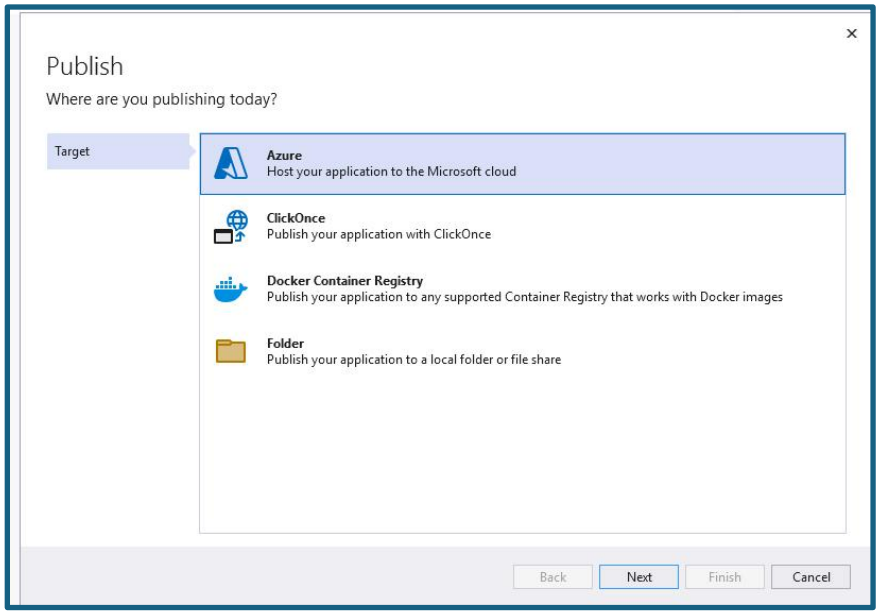
You fire up the wizard by right clicking on your project name over in the project explorer pane. We have seen this menu previously when we were adding files to our project. This time we want to select the publish option. (Figure 24.1)

Figure 24.1: Publish



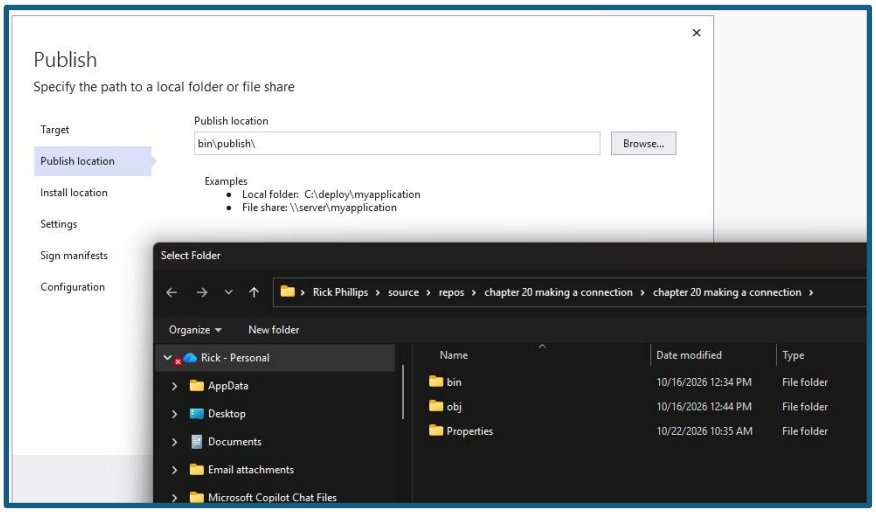
The first wizard screen is all about where you will be publishing to. Azure is Microsoft's cloud, you probably won't be using that until you have several more years of experience. Let's leave Azure to the release professionals. The ClickOnce option is by far the most straightforward and the option we will be choosing. The DockerContainer Registry option is used to package up all the files necessary to run your application. I would choose this option if I were deploying to AWS or Google Cloud to ensure everything I needed was going to be there. Choose the folder option if you already have an instance of this application out there running on IIS and you simply want to update that code. (Figure 24.2)

Figure 24.2: Where to Publish



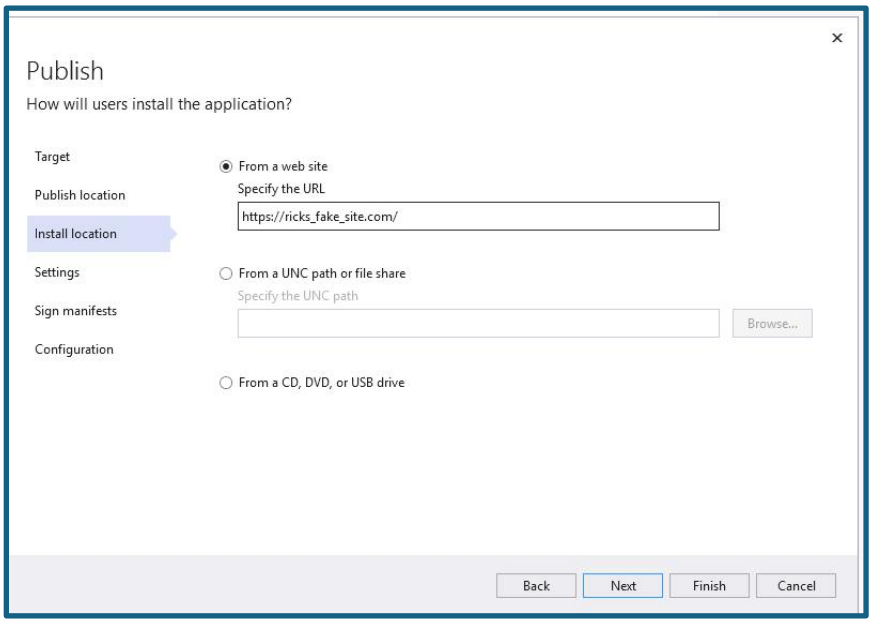
The next wizard screen simply asks you where you want to assemble all these files. As you can see, I have expanded my default loction to show you what Visual Studio is going to pick if you just go with the default. Looks fine to me as this is obviously not where this stuff is ultimately going to end up. This is just a staging location. (Figure 24.3)

Figure 24.3: Staging Location



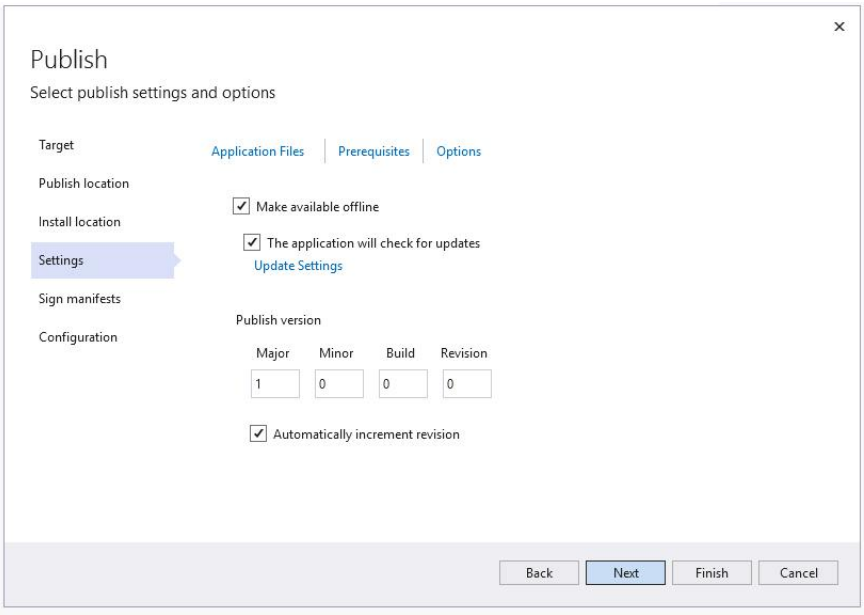
The next wizard page wants to know how users are going to install the application. The typical response is going to be from a web , but you may also want to consider a UNC path. UNC stands for Universal Naming Convention, and you would use a UNC if you wanted everyone in your organization to have access to your app but nobody outside. An example UNC might be \\servername\\sharedfolderpath\\sharedfoldername\\filename(s). The final option is to publish to a CD, DVD or USB drive. Hey it isn't the 1980s anymore, nobody does that today. Microsoft please go ahead and loose that option. (Figure 24.4)

Figure 24.4: Where To Install From



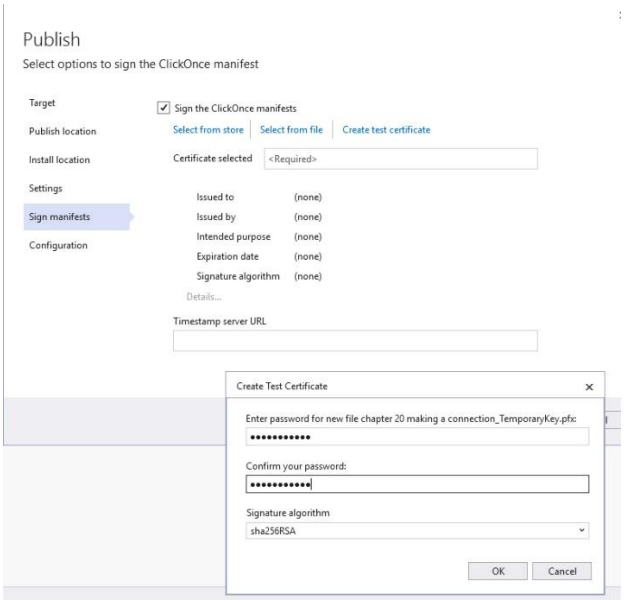
In the next screen you get to determine install settings. Options include setting a version number, setting prerequisites, and making the application available for offline execution. All pretty standard stuff. A typical prerequisite might be a specific version of .NET (Figure 24.5)

Figure 24.5: Install Settings



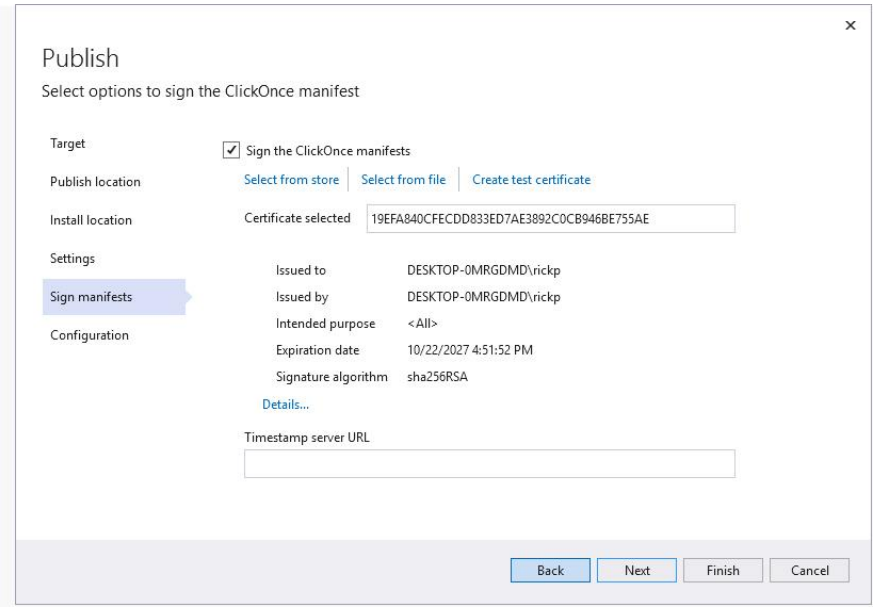
On the next wizard screen you set the application's certificate and password. The signature algorithm will be provided, just go with the default. (Figure 24.6)

Figure 24.6: Certificate



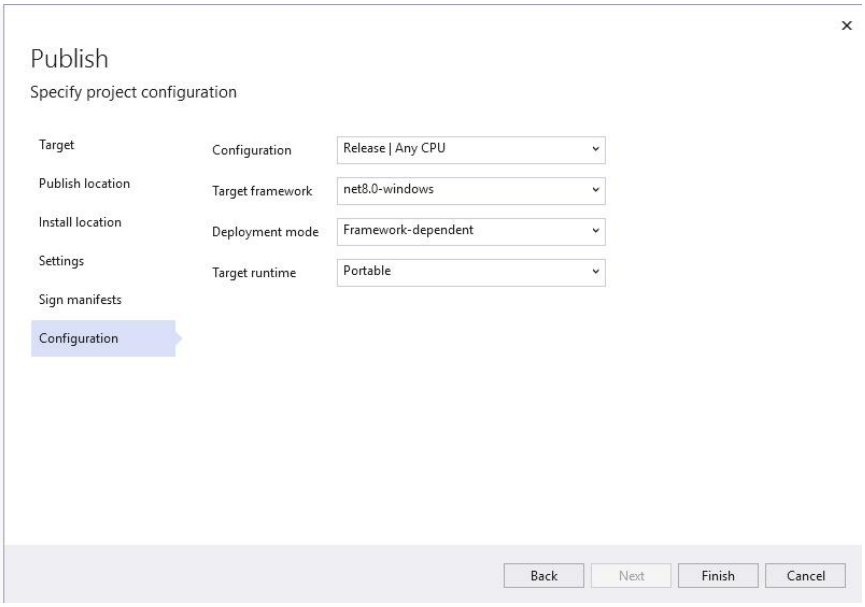
On the next page you set up the URL. Normally you will have to pay a small fee to have this URL registered and activated but don't worry it is just a couple of US dollars. Hopefully your application implements such a good idea that you will recover that amount in just the first few minutes after go-live. (Figure 24.7)

Figure 24.7: URL



The last screen asks you about your project configuration. Make sure to get the Target framework correct as that can become a significant problem. Yes, that is the voice of negative experience talking. (Figure 24.8)

Figure 24.8: Configuration



Okay, that concludes our very short discussion of deployment. As I mentioned at the onset, there are a ton of output options and we just went with the most simple. If you have other specific deployment requirements either work with a release professional or consult the source of all knowledge, ChatGPT.

This also concludes this course. I sincerely hope you enjoyed yourself and are able to benefit from the content. If you thought the course was valuable, please write an appropriate review and I hope to see you in one of my PL/SQL courses.

Finally, go out there and have some fun today. But learn something new today as that is what is really important.